(1.9) Разложение по второй строке

$$\begin{vmatrix} 2 & 3 & -3 & 4 \\ 2 & 1 & -1 & 2 \\ 6 & 2 & 1 & 0 \\ 2 & 3 & 0 & -5 \end{vmatrix} =$$

$$-2\begin{vmatrix} 3 & -3 & 4 \\ 2 & 1 & 0 \\ 3 & 0 & -5 \end{vmatrix} + 1\begin{vmatrix} 2 & -3 & 4 \\ 6 & 1 & 0 \\ 2 & 0 & -5 \end{vmatrix} + 1\begin{vmatrix} 2 & 3 & 4 \\ 6 & 2 & 0 \\ 2 & 3 & -5 \end{vmatrix} + \left(-2\begin{vmatrix} 2 & 3 & -3 \\ 6 & 2 & 1 \\ 2 & 3 & 0 \end{vmatrix}\right)$$

$$\begin{vmatrix} 3 & -3 & 4 \\ 2 & 1 & 0 \\ 3 & 0 & -5 \end{vmatrix} = -2\begin{vmatrix} -3 & 4 \\ 0 & -5 \end{vmatrix} + 1\begin{vmatrix} 3 & 4 \\ 3 & -5 \end{vmatrix} + 0\begin{vmatrix} 3 & -3 \\ 3 & 0 \end{vmatrix}$$

$$-2\begin{vmatrix} -3 & 4 \\ 0 & -5 \end{vmatrix} = (-3 \cdot (-5)) - (0 \cdot 4) = -2(15) = -30$$

$$1\begin{vmatrix} 3 & 4 \\ 3 & -5 \end{vmatrix} = (3 \cdot (-5)) - (3 \cdot 4) = -15 - 12 = -27$$

$$0\begin{vmatrix} 3 & -3 \\ 3 & 0 \end{vmatrix} = (3 \cdot 0) - (3 \cdot (-3)) = 0 \cdot (-9) = 0$$

$$\Delta\begin{vmatrix} 3 & -3 & 4 \\ 2 & 1 & 0 \\ 3 & 0 & -5 \end{vmatrix} = -30 + (-27) + 0 = -57$$

$$\begin{vmatrix} 2 & -3 & 4 \\ 6 & 1 & 0 \\ 2 & 0 & -5 \end{vmatrix} = -6 \begin{vmatrix} -3 & 4 \\ 0 & -5 \end{vmatrix} + 1 \begin{vmatrix} 2 & 4 \\ 2 & -5 \end{vmatrix} + 0 \begin{vmatrix} 2 & -3 \\ 2 & 0 \end{vmatrix}$$

$$(-3 \cdot (-5)) - (0 \cdot 4) = -15 \cdot (-6) = -90$$

$$(2 \cdot (-5)) - (2 \cdot 4) = -10 - 8 = -18$$

$$(2 \cdot 0) - (2 \cdot (-3)) = -6 \cdot 0 = 0$$

$$\Delta \begin{vmatrix} 2 & -3 & 4 \\ 6 & 1 & 0 \\ 2 & 0 & 5 \end{vmatrix} = -90 + (-18) + 0 = \boxed{-108}$$

$$\Delta \begin{vmatrix} 2 & 3 & 4 \\ 6 & 2 & 0 \\ 2 & 3 & -5 \end{vmatrix} = -6 \begin{vmatrix} 3 & 4 \\ 3 & -5 \end{vmatrix} + 2 \begin{vmatrix} 2 & 4 \\ 2 & -5 \end{vmatrix} + 0 \begin{vmatrix} 2 & 3 \\ 2 & 3 \end{vmatrix}$$

$$-6 \cdot \left((3 \cdot (-5)) - (3 \cdot 4) = -15 - 12 = -27 \right) \cdot -6 = 162$$

$$2 \cdot \left((2 \cdot (-5)) - 2 \cdot 4 \right) = -18 (-2) = -36$$

$$0 \left((2 \cdot 3) - (2 \cdot 3) \right) = 0$$

$$\Delta \begin{vmatrix} 2 & 3 & 4 \\ 6 & 2 & 0 \\ 2 & 3 & -5 \end{vmatrix} = 162 + (-36) = \boxed{126}$$

$$\Delta \begin{vmatrix} 2 & 3 & -3 \\ 6 & 2 & 1 \\ 2 & 3 & 0 \end{vmatrix} = -6 \begin{vmatrix} 3 & -3 \\ 3 & 0 \end{vmatrix} + 2 \begin{vmatrix} 2 & -3 \\ 2 & 0 \end{vmatrix} + \left(-1 \begin{vmatrix} 2 & 3 \\ 2 & 3 \end{vmatrix} \right)$$

$$-6\left((3 \cdot 0) - (3 \cdot (-3))\right) = -54$$

$$2\left((2 \cdot 0) - (2 \cdot (-3))\right) = 12$$

$$-1\left((2 \cdot 3) - (2 \cdot 3)\right) = 0$$

$$-54 + 12 + 0 = -42$$

$$\Delta \begin{vmatrix} 2 & 3 & -3 & 4 \\ 2 & 1 & -1 & 2 \\ 6 & 2 & 1 & 0 \\ 2 & 3 & 0 & -5 \end{vmatrix} = -2 \cdot (-57) + 1 \cdot (-108) + 1(126) + 2 \cdot (-42) = 42$$

РАЗЛОЖЕНИЕ ПО 3 СТОЛБЦУ

$$\Delta \begin{vmatrix} 2 & 3 & -3 & 4 \\ 2 & 1 & -1 & 2 \\ 6 & 2 & 1 & 0 \\ 2 & 3 & 0 & -5 \end{vmatrix} =$$

$$-3\begin{vmatrix} 2 & 1 & 2 \\ 6 & 2 & 0 \\ 2 & 3 & -5 \end{vmatrix} + -1\begin{vmatrix} 2 & 3 & 4 \\ 6 & 2 & 0 \\ 2 & 3 & -5 \end{vmatrix} + 1\begin{vmatrix} 2 & 3 & 4 \\ 2 & 1 & 2 \\ 2 & 3 & -5 \end{vmatrix} + 0\begin{vmatrix} 2 & 3 & 4 \\ 2 & 1 & 2 \\ 6 & 2 & 0 \end{vmatrix}$$

$$\Delta \begin{vmatrix} 2 & 1 & 2 \\ 6 & 2 & 0 \\ 2 & 3 & -5 \end{vmatrix} = 2\begin{vmatrix} 6 & 2 \\ 2 & 3 \end{vmatrix} + 0\begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} + -5\begin{vmatrix} 2 & 1 \\ 6 & 2 \end{vmatrix}$$

$$2 \cdot ((6 \cdot 2) - (2 \cdot 2)) = 28$$

$$0 \cdot ((2 \cdot 3) - (2 \cdot 1)) = 0$$

$$-5 \cdot ((2 \cdot 2) - (6 \cdot 1)) = 10$$

$$28 + 0 + 10 = \boxed{38}$$

$$\Delta \begin{vmatrix} 2 & 3 & 4 \\ 6 & 2 & 0 \\ 2 & 3 & -5 \end{vmatrix} = -4\begin{vmatrix} 2 & 3 \\ 6 & 2 \end{vmatrix} + 0\begin{vmatrix} 2 & 3 \\ 2 & 3 \end{vmatrix} + -5\begin{vmatrix} 2 & 3 \\ 6 & 2 \end{vmatrix}$$

$$-4\left((2\cdot 2)-(6\cdot 3)\right) = -4\cdot(-14) = 56$$

$$0\left((2\cdot 3)-(2\cdot 3)\right) = 0$$

$$-5\left((2\cdot 2)-(6\cdot 3)\right) = -5\cdot(-14) = 70$$

$$56 + 0 + 70 = \boxed{126}$$

$$\begin{vmatrix} 2 & 3 & 4 \\ 2 & 1 & 2 \\ 2 & 3 & -5 \end{vmatrix} = 4\begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} + (-2)\begin{vmatrix} 2 & 3 \\ 2 & 3 \end{vmatrix} + -5\begin{vmatrix} 2 & 3 \\ 2 & 1 \end{vmatrix}$$

$$4\left((2\cdot 3)-(2\cdot 1)\right) = 16$$

$$-2\left((2\cdot 3)-(2\cdot 3)\right) = 0$$

$$-5\left((2\cdot 1)-(2\cdot 3)\right) = 20$$

$$16 + 0 + 20 = \boxed{36}$$

$$\Delta = \begin{vmatrix} 2 & 3 & 4 \\ 2 & 1 & -2 \\ 6 & 2 & 0 \end{vmatrix} = 4\begin{vmatrix} 2 & 1 \\ 6 & 2 \end{vmatrix} + (-2)\begin{vmatrix} 2 & 3 \\ 6 & 2 \end{vmatrix} + 0\begin{vmatrix} 2 & 3 \\ 2 & 1 \end{vmatrix}$$

$$4((2\cdot 2)-(6\cdot 1)) = -8$$

$$-2(4-18) = 28$$

$$0((2\cdot 1)-(2\cdot 3)) = 0$$

$$-8+28+0 = \boxed{20}$$

$$\Delta = \begin{vmatrix} 2 & 3 & -3 & 4 \\ 2 & 1 & -1 & 2 \\ 6 & 2 & 1 & 0 \\ 2 & 3 & 0 & -5 \end{vmatrix} = (-3)\cdot 38 - (-1\cdot 126) + 1(36) - 0(70)$$

$$= 48$$

РЕШЕНИЕ СИСТЕМЫ КРАМЕРОМ

$$\begin{cases} x_1 + 2x_2 + x_3 = 8 \\ 4x_1 + 3x_2 - 2x_3 = 4 \\ -x_1 - 2x_2 + x_3 = -2 \end{cases}$$

$$\Delta = \begin{vmatrix} 1 & 2 & 1 \\ 4 & 3 & -2 \\ -1 & -2 & 1 \end{vmatrix} = 1\begin{vmatrix} 3 & -2 \\ -2 & 1 \end{vmatrix} + \left(-2\begin{vmatrix} 4 & -2 \\ -1 & 1 \end{vmatrix}\right) + 1\begin{vmatrix} 4 & 3 \\ -1 & -2 \end{vmatrix}$$

$$1\begin{vmatrix} 3 & -2 \\ -2 & 1 \end{vmatrix} = 1\left((3 \cdot 1) - (-2 \cdot (-2))\right) = 3 - 4 = -1$$

$$-2\begin{vmatrix} 4 & -2 \\ -1 & 1 \end{vmatrix} = -2\left((4 \cdot 1) - (-2 \cdot (-1))\right) = -2(4 - 2) = -4$$

$$1\begin{vmatrix} 4 & 3 \\ -1 & -2 \end{vmatrix} = 1\left((4 \cdot (-2)) - (-1 \cdot 3)\right) = -8 - (-3) = -5$$

$$\Delta = -1 + (-4) + (-5) = -10$$

$$\Delta_1 \begin{vmatrix} 8 & 2 & 1 \\ 4 & 3 & -2 \\ -2 & 2 & 1 \end{vmatrix} = -2\begin{vmatrix} 2 & 1 \\ 3 & -2 \end{vmatrix} + 2\begin{vmatrix} 8 & 1 \\ 4 & -2 \end{vmatrix} + 1\begin{vmatrix} 8 & 2 \\ 4 & 3 \end{vmatrix}$$

$$-2\left((2 \cdot (-2)) - (3 \cdot 1)\right) = -2(-4 - 3) = 14$$

$$2\left((8 \cdot (-2)) - (4 \cdot 1)\right) = 2(-16 - 4) = -40$$

$1((8\cdot 3)-(4\cdot 2)) = 24-8 = 16$

$\Delta_1 = 14 + (-40) + 16 = \boxed{-10}$

$$\Delta_2 = \begin{vmatrix} 1 & 8 & 1 \\ 4 & 4 & -2 \\ -1 & 2 & 1 \end{vmatrix} = -1\begin{vmatrix} 8 & 1 \\ 4 & -2 \end{vmatrix} + 2\begin{vmatrix} 1 & 1 \\ 4 & -2 \end{vmatrix} + 1\begin{vmatrix} 1 & 8 \\ 4 & 4 \end{vmatrix}$$

$-1((8\cdot(-2))-4) = -1(-16-4) = 20$

$2((4\cdot(-2))-4) = 2(-2-4) = -12$

$1((1\cdot 4)-(4\cdot 8)) = 4-32 = -28$

$\Delta_2 = 20 + (-12) + (-28) = \boxed{-20}$

$$\Delta_3 = \begin{vmatrix} 1 & 2 & 8 \\ 4 & 3 & 4 \\ -1 & 2 & 2 \end{vmatrix} = -1\begin{vmatrix} 2 & 8 \\ 3 & 4 \end{vmatrix} + 2\begin{vmatrix} 1 & 8 \\ 4 & 4 \end{vmatrix} + \left(-2\begin{vmatrix} 1 & 2 \\ 4 & 3 \end{vmatrix}\right)$$

$-1((2\cdot 4)-(3\cdot 8)) = -1(8-24) = 16$

$2((1\cdot 4)-(4\cdot 8)) = 2(4-32) = -56$

$-2((1\cdot 3)-(4\cdot 2)) = -2(3-8) = 10$

$\Delta_3 = 16 + (-56) + 10 = \boxed{-30}$

$$x_1 = \frac{-10}{-10} = 1$$

$$x_2 = \frac{-20}{-10} = 2$$

$$x_3 = \frac{-30}{-10} = 3$$

$1 + 2\cdot 2 + 3 = 8$

$4\cdot 1 + 3\cdot 2 - 2\cdot 3 = 4$

$-1 - 2\cdot 2 + 3 = -2$